ANDROID PROGRAMMING PRACTICAL 03

Module 1 · Activity Lifecycle and Logcat

Aim

Demonstrate Activity lifecycle callbacks and analyze execution using Logcat.

Project name: ActivityLifecycleDemo

Requirements

Android Studio, Java, Android SDK, and an Android Virtual Device or compatible Android phone. Use the package name generated by your project and adjust imports/dependencies to the installed Android Studio template.

Procedure

1. Create an Empty Views Activity project.

2. In MainActivity.java, override onCreate(), onStart(), onResume(), onPause(), onStop(), and onDestroy().

3. Call the matching superclass method in every callback.

4. Add Log.d("ActivityLifecycle", "onCreate called")-style messages in each callback.

5. Run the app and open Logcat; filter by the ActivityLifecycle tag.

6. Launch the activity, press Home, return to the app, rotate the device if supported, and close the activity.

7. Record the callback order observed; note that exact sequences can vary with device and configuration.

Implementation checklist

• Keep resource names lowercase and use unique view IDs.

• Make sure every referenced layout, menu, drawable, string, or Java class exists in the project.

• Build the project and resolve compile errors before running it.

• Use Logcat to diagnose runtime errors and permissions or manifest issues.

Expected result

Lifecycle callbacks are recorded in Logcat as the activity changes state.

Testing / observations

Record the device/API level, the actions tested, the observed output, and any errors corrected. Attach a screenshot of the successful run to your practical record if required by your institution.

Conclusion

The stated Android concept was implemented and verified through the steps and tests above.